THQ #6

By: Ethan Le :D

Question 1 (Section 5.2)

Objective Function:

$$\textbf{min } -2x_1 - 3x_2 - 3x_3$$

Constraints:

- $3x_1 + 2x_2 + x_4 = 60$
- $-x_1 + x_2 + 4x_3 + x_5 = 10$
- $2x_1 - 2x_2 + 5x_3 + x_6 = 50$
- $x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$ **(non-negative)**

Given Tableaux:

Section 5.2:

Consider the optimization problem below and the related initial and final tableaux.

	x_1	x_2	x_3	x_4	x_5	x_6	
x_4	3	2	0	1	0	0	60
x_5	−1	(1)	4	0	1	0	10
x_6	2	−2	5	0	0	1	50
	−2	−3	−3	0	0	0	0
x_4	(5)	0	−8	1	−2	0	40
x_2	−1	1	4	0	1	0	10
x_6	0	0	13	0	2	1	70
	−5	0	9	0	3	0	30
x_1	1	0	$-\frac{8}{5}$	$\frac{1}{5}$	$-\frac{2}{5}$	0	8
x_2	0	1	$\frac{12}{5}$	$\frac{1}{5}$	$\frac{3}{5}$	0	18
x_6	0	0	13	0	2	1	70
	0	0	1	1	1	0	70

Minimize $-2x_1 - 3x_2 - 3x_3$

subject to

$$\begin{aligned} 3x_1 + 2x_2 + x_4 &= 60 \\ -x_1 + x_2 + 4x_3 + x_5 &= 10 \\ 2x_1 - 2x_2 + 5x_3 + x_6 &= 50 \\ x_1, x_2, x_3, x_4, x_5, x_6 &\geq 0 \end{aligned}$$

1. **Determine B_0, B, b, b*, c, c***
2. **Determine A_B, A^*_{B0}, c_B , and A^{-1}_B**
3. **Check that** $b^* = A_B^{-1} b$ **and** $z^* = z_0 + c_B b^*$

Figure 1: Initial, intermediate, and final tableaux

Tasks:

1. Determine B_0, B, b, b^*, c, c^*.

$$B_0 = \{x_4, x_5, x_6\}, \qquad B = \{x_1, x_2, x_6\}$$

$$b = [60,\ 10,\ 50]^\top, \qquad b^* = [8,\ 18,\ 70]^\top$$

c and c^* are the bottom rows of the initial and final tableaux respectively:

$$c^\top = [-2,\ -3,\ -3,\ 0,\ 0,\ 0], \qquad {c^*}^\top = [0,\ 0,\ 1,\ 1,\ 1,\ 0]$$

2. Determine A_B, $A^*_{B_0}$, c_B, and A_B^{-1}.

A_B = columns of $B = \{x_1, x_2, x_6\}$ in the initial tableau:

$$A_B = \begin{bmatrix} 3 & 2 & 0 \\ -1 & 1 & 0 \\ 2 & -2 & 1 \end{bmatrix}$$

$A_B^{-1} = A^*_{B_0}$ = columns of $B_0 = \{x_4, x_5, x_6\}$ in the final tableau:

$$A_B^{-1} = A^*_{B_0} = \begin{bmatrix} \frac{1}{5} & -\frac{2}{5} & 0 \\ \frac{1}{5} & \frac{3}{5} & 0 \\ 0 & 2 & 1 \end{bmatrix}$$

c_B = the c-row entries for just the B variables $\{x_1, x_2, x_6\}$:

$$c_B = [-2,\ -3,\ 0]^\top$$

3. Check that $b^* = A_B^{-1} b$ and $z^* = z_0 + c_B b^*$.

$$A_B^{-1} b = \begin{bmatrix} \frac{1}{5} & -\frac{2}{5} & 0 \\ \frac{1}{5} & \frac{3}{5} & 0 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 60 \\ 10 \\ 50 \end{bmatrix} = \begin{bmatrix} 12 - 4 \\ 12 + 6 \\ 20 + 50 \end{bmatrix} = \begin{bmatrix} 8 \\ 18 \\ 70 \end{bmatrix} = b^*$$

$$z_0 + c_B^\top b^* = 0 + [-2,\ -3,\ 0] \begin{bmatrix} 8 \\ 18 \\ 70 \end{bmatrix} = -16 - 54 = -70 = z^*$$

Question 2 (Section 5.1 #15)

A company produces units of A, B, and C using raw materials M_1, M_2, and labor.

	A	B	C	Available
M_1 (lb)	6	12	4	1 ton = 2000 lb
M_2 (lb)	16	25	7	2.5 tons = 5000 lb
Labor (hr)	2	3	1	500 hr at \$18/hr
Profit (\$)	105	165	60	

Up to 120 overtime hours are available at \$24/hr.

Variables: x_1, x_2, x_3 = units of A, B, C; x_4 = overtime hours used.

Objective Function:

$$\textbf{max } z = 105x_1 + 165x_2 + 60x_3 - 18(2x_1 + 3x_2 + x_3) - 6x_4 = 69x_1 + 111x_2 + 42x_3 - 6x_4$$

Constraints:

- $6x_1 + 12x_2 + 4x_3 \leq 2000$
- $16x_1 + 25x_2 + 7x_3 \leq 5000$
- $2x_1 + 3x_2 + x_3 \leq 500 + x_4$
- $x_4 \leq 120$
- $x_1, x_2, x_3, x_4 \geq 0$ **(non-negative)**

Given Tableaux (Table 5.2):

Tasks:

(a) What is the optimal production schedule and profit?

From the final tableau, the basic variables are x_1, x_3, x_4 (and slack x_6), with x_2 nonbasic, so

$$(x_1, x_2, x_3, x_4) = (240,\ 0,\ 140,\ 120), \qquad z^* = 21720.$$

(b) Write the dual problem and determine the optimal solution point.

Rewrite the labor constraint as $2x_1+3x_2+x_3-x_4 \leq 500$. The primal is a maximization with all $\leq$ constraints and $x_j \geq 0$, so the dual is a minimization with all $\geq$ constraints and $y_i \geq 0$.

Dual Objective Function:

$$\textbf{min } 2000y_1 + 5000y_2 + 500y_3 + 120y_4$$

Dual Constraints:

- $6y_1 + 16y_2 + 2y_3 \geq 69$
- $12y_1 + 25y_2 + 3y_3 \geq 111$
- $4y_1 + 7y_2 + y_3 \geq 42$
- $-y_3 + y_4 \geq -6$
- $y_1, y_2, y_3, y_4 \geq 0$ **(non-negative)**

The optimal dual point is read from the bottom row of the final primal tableau under the slack columns x_5, x_6, x_7, x_8:

$$(y_1, y_2, y_3, y_4) = (7.5,\ 0,\ 12,\ 6).$$

Check against the dual objective (should match $z^* = 21720$):

$$2000(7.5) + 5000(0) + 500(12) + 120(6) = 15000 + 0 + 6000 + 720 = 21720.$$

(c) Should the manager accept an offer for more overtime hours at \$24/hr?

More overtime means raising the cap $x_4 \leq 120$. That constraint's slack is x_8, so its shadow price is the bottom-row entry under x_8, which is $y_4 = 6$.

Since $y_4 = 6 > 0$, one more overtime hour at \$24/hr (already in the model) increases profit by about \$6. So yes, the manager should accept.

(d) Should additional M_1 be purchased at a cost of \$7.75/lb over the current price?

M_1 is the first constraint, with slack x_5, so its shadow price is $y_1 = 7.5$. One more pound of M_1 is only worth about \$7.50, but the extra cost is \$7.75 $>$ 7.50. So no, do not buy additional M_1.

Section 5.1 #15

15. Combining raw materials M_1 and M_2 and labor, a company produces units of A, B, and C. The requirements and profit (excluding the cost of labor) for the production and sale of a unit of each are as follows:

	M_1 (lb)	M_2 (lb)	Labor (hr)	Profit ($)
A	6	16	2	105
B	12	25	3	165
C	4	7	1	60

For the next month, the company has available 1 ton of M_1, 2.5 tons of M_2, 500 hr of labor at \$18/hr, and up to an additional 120 hr of overtime at \$24/hr. (The company pays only for the labor used.)

To determine an optimal production schedule, the company manager defines x_1, x_2, and x_3 to be the number of A's, B's, and C's to be produced and x_4 to be the number of hours of overtime to be used and formulates the following model:

$$\begin{aligned}\text{Maximize } z &= 105x_1 + 165x_2 + 60x_3 - 18(2x_1 + 3x_2 + x_3) - 6x_4 \\ &= 69x_1 + 111x_2 + 42x_3 - 6x_4\end{aligned}$$

subject to

$$\begin{aligned} 6x_1 + 12x_2 + 4x_3 &\le 2000 \\ 16x_1 + 25x_2 + 7x_3 &\le 5000 \\ 2x_1 + 3x_2 + x_3 &\le 500 + x_4 \\ x_4 &\le 120 \end{aligned}$$

$$x_1, x_2, x_3, x_4 \ge 0$$

Adding four slack variables and applying the simplex algorithm, the reduced tableaux resolution (initial and final tableaux only) are shown in Table 5.2.

(a) What is the optimal production schedule, and what profit does it yield?
(b) Write out the dual problem and determine an optimal solution point.
(c) Several employees offer to work additional hours of overtime (at the same \$24/hr rate). Should the manager accept their offer?
(d) Suppose additional pounds of M_1 could be purchased, at a cost of \$7.75/lb over what the company now pays for the raw material. Should more be purchased?

Table 5.2

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	
x_5	6	12	4	0	1	0	0	0	2000
x_6	16	25	7	0	0	1	0	0	5000
x_7	2	3	1	−1	0	0	1	0	500
x_8	0	0	0	1	0	0	0	1	120
	−69	−111	−42	6	0	0	0	0	0
x_6	0	4	0	0	1	1	−11	−11	180
x_4	0	0	0	1	0	0	0	1	120
x_1	1	0	0	0	$-\frac{1}{2}$	0	2	2	240
x_3	0	3	1	0	1	0	−3	−3	140
	0	15	0	0	$7\frac{1}{2}$	0	12	6	21,720

Figure 2: Initial and final tableaux for Question 2

Question 3 (Section 5.3 #9a & c)

Same LP and tableaux as Question 2 (Section 5.1 #15).

Important info:

Initial basis (ordered to form the identity in A):

$$B_0 = [x_5,\ x_6,\ x_7,\ x_8]$$

Final basis (ordered to form the identity in A^*):

$$B = [x_6,\ x_4,\ x_1,\ x_3]$$

Tasks:

(a) Determine how much the profit on a unit of A (the \$105) can vary before the optimal production schedule would change.

Since \$105 correlates with the net objective coefficient 69 for x_1, we replace 69 by $69 + \varepsilon$.

c_λ is the z-row of the initial tableau with that change (the x_1 entry becomes $-69 - \varepsilon$):

$$c_\lambda^\top = [-69 - \varepsilon,\ -111,\ -42,\ 6,\ 0,\ 0,\ 0,\ 0]$$

$$c_{\lambda B}^\top = [0,\ 6,\ -69 - \varepsilon,\ -42]$$

$$c_\lambda^* = c_\lambda - c_{\lambda B} A^*$$

$$= \begin{bmatrix} -69-\varepsilon & -111 & -42 & 6 & 0 & 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 6 & -69-\varepsilon & -42 \end{bmatrix} \begin{bmatrix} 0 & 4 & 0 & 0 & 1 & 1 & -11 & -11 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 2 & 2 \\ 0 & 3 & 1 & 0 & 1 & 0 & -3 & -3 \end{bmatrix}$$

Matrix multiplication $c_{\lambda B} A^*$:

$$c_{\lambda B} A^* = [-69 - \varepsilon,\ -126,\ -42,\ 6,\ -7.5 + \tfrac{1}{2}\varepsilon,\ 0,\ -12 - 2\varepsilon,\ -6 - 2\varepsilon]$$

Subtract from c_λ:

$$c_\lambda^* = [0,\ 15,\ 0,\ 0,\ 7.5 - \tfrac{1}{2}\varepsilon,\ 0,\ 12 + 2\varepsilon,\ 6 + 2\varepsilon]$$

Keep every entry ≥ 0:

- $7.5 - \dfrac{1}{2}\varepsilon \geq 0 \implies \varepsilon \leq 15$
- $12 + 2\varepsilon \geq 0 \implies \varepsilon \geq -6$
- $6 + 2\varepsilon \geq 0 \implies \varepsilon \geq -3$

The binding bounds are $-3 \le \varepsilon \le 15$, so the sale profit on A can range from

$$\$102 \le 105 + \varepsilon \le \$120$$

before the optimal production schedule changes.

(c) Determine how much the profit on a unit of C (the \$60) can vary before the optimal production schedule would change.

Since \$60 correlates with the net objective coefficient 42 for x_3, we replace 42 by $42 + \varepsilon$.

c_λ is the z-row of the initial tableau with that change (the x_3 entry becomes $-42 - \varepsilon$):

$$c_\lambda^\top = [-69,\ -111,\ -42 - \varepsilon,\ 6,\ 0,\ 0,\ 0,\ 0]$$

$c_{\lambda B}$ is the same row, but only the final basis columns $B = [x_6,\ x_4,\ x_1,\ x_3]$:

$$c_{\lambda B}^\top = [0,\ 6,\ -69,\ -42 - \varepsilon]$$

$$c_\lambda^* = c_\lambda - c_{\lambda B} A^*$$

$$= \begin{bmatrix} -69 & -111 & -42-\varepsilon & 6 & 0 & 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 6 & -69 & -42-\varepsilon \end{bmatrix} \begin{bmatrix} 0 & 4 & 0 & 0 & 1 & 1 & -11 & -11 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 2 & 2 \\ 0 & 3 & 1 & 0 & 1 & 0 & -3 & -3 \end{bmatrix}$$

Matrix multiplication $c_{\lambda B} A^*$:

$$c_{\lambda B} A^* = [-69,\ -126 - 3\varepsilon,\ -42 - \varepsilon,\ 6,\ -7.5 - \varepsilon,\ 0,\ -12 + 3\varepsilon,\ -6 + 3\varepsilon]$$

Subtract from c_λ:

$$c_\lambda^* = [0,\ 15 + 3\varepsilon,\ 0,\ 0,\ 7.5 + \varepsilon,\ 0,\ 12 - 3\varepsilon,\ 6 - 3\varepsilon]$$

Keep every entry ≥ 0:

- $15 + 3\varepsilon \ge 0 \implies \varepsilon \ge -5$
- $7.5 + \varepsilon \ge 0 \implies \varepsilon \ge -7.5$
- $12 - 3\varepsilon \ge 0 \implies \varepsilon \le 4$
- $6 - 3\varepsilon \ge 0 \implies \varepsilon \le 2$

The binding bounds are $-5 \le \varepsilon \le 2$, so the sale profit on C can range from

$$\$55 \le 60 + \varepsilon \le \$62$$

before the optimal production schedule changes.

Question 4 (Section 5.5 #7)

Same LP and tableaux as Question 2 (Section 5.1 #15).

Important info:

$$B_0 = [x_5,\ x_6,\ x_7,\ x_8], \qquad B = [x_6,\ x_4,\ x_1,\ x_3]$$

$$A_B^{-1} = A_{B_0}^* = \text{final-tableau columns under } x_5, x_6, x_7, x_8$$

These are **RHS / b-changes** (handout page 5: "For λ change in RHS $\rightarrow$ b-entries"), not objective c-changes like Question 3.

Tasks:

(a) Suppose $(500 + \lambda)$ hr of labor at \$18/hr can be used, with $-16 \le \lambda \le 16$, and no other data change. Determine the optimal production schedule and profit.

Method: 1. Replace the labor RHS by $500 + \lambda$, so

$$b_\lambda = [2000,\ 5000,\ 500 + \lambda,\ 120]^\top.$$

2. The given interval means the current basis stays optimal, so recompute

$$b_\lambda^* = A_B^{-1} b_\lambda.$$

From the final tableau, columns under x_5, x_6, x_7, x_8 (rows ordered as $B = [x_6, x_4, x_1, x_3]$):

$$A_B^{-1} = \begin{bmatrix} 1 & 1 & -11 & -11 \\ 0 & 0 & 0 & 1 \\ -\frac{1}{2} & 0 & 2 & 2 \\ 1 & 0 & -3 & -3 \end{bmatrix}$$

$$b_\lambda^* = \begin{bmatrix} 1 & 1 & -11 & -11 \\ 0 & 0 & 0 & 1 \\ -\frac{1}{2} & 0 & 2 & 2 \\ 1 & 0 & -3 & -3 \end{bmatrix} \begin{bmatrix} 2000 \\ 5000 \\ 500 + \lambda \\ 120 \end{bmatrix} = \begin{bmatrix} 1(2000) + 1(5000) - 11(500 + \lambda) - 11(120) \\ 120 \\ -\frac{1}{2}(2000) + 2(500 + \lambda) + 2(120) \\ 1(2000) - 3(500 + \lambda) - 3(120) \end{bmatrix} = \begin{bmatrix} 180 - 11\lambda \\ 120 \\ 240 + 2\lambda \\ 140 - 3\lambda \end{bmatrix}$$

So with basics ordered $[x_6, x_4, x_1, x_3]$:

$$b_\lambda^* = [180 - 11\lambda,\ 120,\ 240 + 2\lambda,\ 140 - 3\lambda]^\top.$$

3. Read the new production schedule from b_λ^* (same basic variables x_6, x_4, x_1, x_3; nonbasics stay 0).

$$(x_1, x_2, x_3, x_4) = (240 + 2\lambda,\ 0,\ 140 - 3\lambda,\ 120)$$

(valid for $-16 \le \lambda \le 16$).

4. Get the new profit from $z_\lambda^* = z_0 + c_B^\top b_\lambda^*$.

Using the max-form costs for $B = [x_6, x_4, x_1, x_3]$:

$$c_B^\top = [0,\ -6,\ 69,\ 42], \qquad z_0 = 0$$

$$
\begin{aligned}
z^*_\lambda &= [0,\ -6,\ 69,\ 42] \begin{bmatrix} 180 - 11\lambda \\ 120 \\ 240 + 2\lambda \\ 140 - 3\lambda \end{bmatrix} \\
&= -6(120) + 69(240 + 2\lambda) + 42(140 - 3\lambda) \\
&= -720 + 16560 + 138\lambda + 5880 - 126\lambda \\
&= 21720 + 12\lambda.
\end{aligned}
$$

Check: plug the schedule into the max objective:

$$69(240 + 2\lambda) + 42(140 - 3\lambda) - 6(120) = 21720 + 12\lambda.$$

At $\lambda = 0$, this is 21720, matching the original final tableau.

(b) With 1 ton of M_1 available, its marginal value is \$7.50/lb. For what interval around 2000 lb does M_1 keep this marginal value?

Marginal value = shadow price $y_1 = 7.5$ (slack x_5 bottom-row entry). Since R_1 (the M_1 constraint) correlates to slack x_5, replace the M_1 RHS by $2000 + \varepsilon$:

$$b_\varepsilon = [2000 + \varepsilon,\ 5000,\ 500,\ 120]^\top.$$

Compute $b^*_\varepsilon = A_B^{-1} b_\varepsilon$ and require every entry ≥ 0.

$$
b^*_\varepsilon = \begin{bmatrix} 1 & 1 & -11 & -11 \\ 0 & 0 & 0 & 1 \\ -\frac{1}{2} & 0 & 2 & 2 \\ 1 & 0 & -3 & -3 \end{bmatrix} \begin{bmatrix} 2000 + \varepsilon \\ 5000 \\ 500 \\ 120 \end{bmatrix} = \begin{bmatrix} 1(2000 + \varepsilon) + 1(5000) - 11(500) - 11(120) \\ 120 \\ -\frac{1}{2}(2000 + \varepsilon) + 2(500) + 2(120) \\ 1(2000 + \varepsilon) - 3(500) - 3(120) \end{bmatrix} = \begin{bmatrix} 180 + \varepsilon \\ 120 \\ 240 - \frac{1}{2}\varepsilon \\ 140 + \varepsilon \end{bmatrix}
$$

Require $b^*_\varepsilon \geq 0$:

- $180 + \varepsilon \geq 0 \implies \varepsilon \geq -180$
- $120 \geq 0$ always
- $240 - \frac{1}{2}\varepsilon \geq 0 \implies \varepsilon \leq 480$
- $140 + \varepsilon \geq 0 \implies \varepsilon \geq -140$

Binding bounds: $-140 \leq \varepsilon \leq 480$.

That ε-interval is the range of M_1 availability where the \$7.50/lb shadow price remains valid:

$$1860 \leq 2000 + \varepsilon \leq 2480 \text{ lb of } M_1.$$